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Ashley Muddiman, Shannon C. McGregor & Natalie Jomini Stroud

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(Re)Claiming Our Expertise: Parsing Large Text Corpora With Manually Validated and Organic Dictionaries

ASHLEY MUDDIMAN, SHANNON C. MCGREGOR, and NATALIE JOMINI STROUD

Content analysis of large-scale textual data sets poses myriad problems, particularly when researchers seek to analyze content that is both theoretically derived and context dependent. In this piece, we detail the approach we developed to tackle the analysis of the context-dependent content of political incivility. After describing our manually validated organic dictionaries approach, we compare the method to others we could have used and then replicate the method in a different—but still context-dependent—project examining political issue content on social media. We conclude by summarizing the strengths and weaknesses of the approach and offering suggestions for future research that can refine and expand the method.

Keywords computer-aided content analysis, incivility, news comments, Twitter, news issues

When conducting a content analysis of a large data set, researchers have a number of tools at their disposal (Burscher, Vliegenthart, & De Vreese, 2015; Hopkins, & King, 2010; Young & Soroka, 2012); yet, as others have noted, there are limitations with current computational approaches (Carlson & Montgomery, 2017; Grimmer & Stewart, 2013; Guo, Vargo, Pan, Ding, & Ishwar, 2016). Human coders cannot reasonably code hundreds of thousands or millions of texts, and a sample of manageable size does not scratch the surface of the content (Grimmer & Stewart, 2013; Guo et al., 2016). We add a new technique to the large-scale content analysis toolbox—manually validated organic dictionaries. Our method forefronts a context-dependent understanding of language, allowing political communication scholars to achieve theoretically nuanced analysis of complex concepts.

In this article, we detail our approach to tackling the analysis of the context-dependent content of political incivility. After describing the manually validated organic

Ashley Muddiman is an assistant professor in the Department of Communication Studies, University of Kansas. Shannon C. McGregor is an assistant professor in the Department of Communication, University of Utah. Natalie Jomini Stroud is an associate professor in the Department of Communication Studies and the School of Journalism and director of the Center for Media Engagement, University of Texas at Austin.

Address correspondence to Ashley Muddiman, Department of Communication Studies, University of Kansas, Bailey Hall, Room 102, 1440 Jayhawk Blvd., Lawrence, KS 66045. E-mail: ashley.muddiman@ku.edu

dictionaries approach, we compare the method to others we could have used and then replicate the method in a different—but still context-dependent—project examining political issue content on social media. We conclude by summarizing the strengths and weaknesses of the manually validated organic dictionary approach, and offering suggestions for future research that can refine and expand the method.

Incivility: A Theoretical Construct That Varies by Context

In this article, we provide a case study focusing on uncivil content in news comments to demonstrate the usefulness of the manually validated organic dictionary approach. As a first step, it is important to define incivility—something that is tricky even at the conceptual level (Coe, Kenski, & Rains, 2014; Maisel, 2012). Although some researchers define incivility as extreme negativity (Fridkin & Kenney, 2008), for this project, and consistent with other scholars, we conceptualized incivility as violations of both politeness norms (e.g., name-calling and swearing; Mutz, 2015) and democratic norms (e.g., claims of discrimination, government dysfunction, and treason; Muddiman, 2017; Papacharissi, 2004).

Although theoretically well-developed in the literature, such a definition poses operational challenges. Even with data sets of a manageable size for manual coding, categorizing incivility reliably across human coders is challenging (Coe et al., 2014; Stroud, Scacco, Muddiman, & Curry, 2015). Incivility coding with computer-aided methods on a large-scale data set presents even greater challenges. First, although a number of preconstructed, out-of-the-box dictionary approaches exist to measure negative language, they do not necessarily align with this theoretical approach to incivility. For example, words like "critic" and "doubt," which appear in the popular Lexicoder sentiment dictionary, may be normatively beneficial in the context of political discussion. Alternatively, "unconstitutional" does not appear in the Lexicoder sentiment dictionary, yet it could indicate a violation of democratic norms (see Young & Soroka, 2012).

Second, several previous studies that have investigated incivility on a large scale have approached incivility more narrowly; for instance, examining a mixture of impoliteness and moral words (Theocharis, Barberá, Fazekas, Popa, & Parnet, 2016) or personal attacks (Wulczyn, Thain, & Dixon, 2017). Our conceptualization included a wider spectrum of norm violations, pushing us to develop a method that could account for various theoretical approaches to incivility.

Finally, incivility norms are context dependent, representing the customs and traditions generated within cultures and situations (Aarts & Dijksterhuis, 2003; Sherif, 1936). Take, for example, the word "devil," which would be considered an insult if it were used to described a political candidate. However, calling someone a "Blue Devil" in the context of sports references Duke University's mascot rather than an insult. To capture incivility, broadly defined, we had to balance competing goals: keeping in mind both theoretical conceptualizations of uncivil norm violations *and* the context from which the incivility emerged.

Manually Validated Organic Dictionaries

To analyze uncivil language in comments from *The New York Times*, we developed a method of constructing and validating content dictionaries that includes human checks and inductive and deductive processes.[1] As researchers have noted, standard practices of textual data analysis are in flux (Shah, Cappella, & Neuman, 2015), meaning that we had little guidance as to the best way to go about this analysis. The method we developed is detailed next.

Creating a Top-Features List

We first created a list of the top words used in our data set of news comments, following previous research (e.g., Jurafsky & Martin, 2016; Pang, Lee, & Vaithyanathan, 2002; Schwartz & Ungar, 2015). Starting with a list of terms from the data set of interest is important in the case of context-dependent variables, like incivility.

This approach can be distinguished from other techniques we could have employed. Viewing the frequently used words in the data set allowed us to familiarize ourselves with the language of the *Times* commenting community rather than attempting to create a dictionary from scratch or using an out-of-the-box dictionary created by others. Another approach, supervised machine learning, requires a coded sample of data to be used as a training data set for the development of an algorithm (Burscher et al., 2015). Typically, a small sample relative to the size of the large data set is used for training. It is conceivable that some of the most-frequently appearing words in the entire data set may *not* appear in the sample because language use is diverse and even oft-used words may not appear frequently enough to be captured in a sample. Theocharis and colleagues (2016) provide a good example of this. In one of the countries they studied, they collected more than 400,000 tweets and then sampled 7,000 of those tweets (approximately 2% of the sample) for a training set. Of those 7,000 tweets, only 121 (2%) included impoliteness. Since uncivil words were infrequent in our data set as well—the most used word related to incivility (threat, 81,952 comments) appeared in less than 1% of all comments—randomly selecting comments for a training set would have increased the likelihood of missing even these most frequently used terms. We share this example not to quibble with Theocharis and colleagues' findings, but rather to point out a rationale for focusing on the words used most often in a particular data set.

We removed stop words (e.g., "the", "a"), then used the Porter (1980) stemmer to remove the endings of the words and combine similar word roots to create "features" (e.g., "mislead" and "misleading" were captured with the feature "mislead").[2,3] We ran the Natural Language Toolkit program (Bird, Klein, & Loper, 2009) in Python to compile the 5,000 features used most often in the data set. This is similar to previous research, although there is no standard cutoff for the most frequently used features list (see, for instance, Vargo, Guo, McCombs, & Shaw [2014], who listed words that appeared more than 1,000 times). Our top features appeared in between 6,296 ("censu") and 1,854,559 ("people") comments.

Selecting Features to Test

Next, we brought a deductive approach to the most-used features list. Previous research, notably Coe and colleagues (2014), Muddiman (2017), Papacharissi (2004), and Stroud and colleagues (2015), guided us as we created a list of features potentially related to theoretical violations of politeness (e.g., name-calling) and democratic (e.g., discrimination, political threats, dysfunction, misinformation) norms.

Two of the co-authors repeatedly reviewed the top-features list and selected any feature that could be related to the focus of the dictionary. We over-selected the features, choosing those that were very likely related (e.g., "dumb") *and* those that had a much smaller chance at being related (e.g., "crimin"). Thinking broadly at this step was important to ensure that no features potentially related to the dictionary were ignored. It seemed better to over-select features that ended up not being included in the dictionaries than to under-select features that should have been included. We selected 88 features potentially related to incivility.

Manual Validation of Dictionaries

Our next step was the most important—and novel—part of the process: human validation of each of the potential dictionary features. Creating final dictionaries based only on researcher-selected features would face a major validity threat: the features in the list would be entirely out of context and may not be used in a manner appropriate for the dictionary. Most projects that use a dictionary approach either do not validate the dictionaries or validate the dictionaries as a whole rather than each dictionary term (Ksiazek, Peer, & Zivic, 2015; Neuman, Guggenheim, Jang, & Bae, 2014; Vargo et al., 2014). We suggest a middle step: each feature included in a dictionary should be validated by human coders to ensure that the texts utilize the features in a way that aligns with the purpose of the dictionary.

Human Coders. Two human coders read the comments for each feature and independently determined whether or not the feature should be included in the dictionary. We used five graduate student (MA and PhD level) and faculty coders to judge whether each feature was used in a manner related to incivility (see Appendix A for codebook). We used trained, as opposed to amateur, coders because the ultimate goal of our study was to examine how news users and journalists engage with comment content that has been theoretically conceptualized as uncivil in past research. We note that amateur coders could be useful for testing how individuals perceive incivility (e.g., Kenski, Coe, & Rains, in press). If perceived incivility were the goal, however, we would need to use a representative sample since research shows that perceptions of incivility vary across individuals (Kenski et al., in press; Muddiman, 2017; Mutz, 2015). Getting a representative sample to evaluate sufficient numbers of comments was not feasible. Overall, using trained coders allowed us to create theoretically driven dictionaries.

Coding the Features. To conduct the manual validation test of the features, we randomly chose 25 comments for each of the 88 potential incivility features, totaling 2,200 comments for human coding. This number of comments accomplished two goals: providing a signal for each feature as to whether it should be included in the final dictionary, while producing a manageable number of comments to code manually.

Importantly, the coding unit for this reliability process was the *feature* rather than the *comment*. If, for instance, coders were examining the feature “illeg,” they were told to examine only words like “illegal” or “illegally” in the comment. If “illegal immigration” was mentioned without disparaging someone, but earlier in a comment President Barack Obama was called “a divisive race-baiting community organizer,” the coder would still report that the comment did not use “illeg” in an uncivil way. This coding unit was chosen for two reasons. First, conceptually, the end goal was to decide whether to include a feature in a dictionary, not whether each comment included words related to incivility. As the “illegal” example illustrates, we observed instances in our data where the feature we analyzed was *not* used uncivilly, but the comment contained other words used in uncivil ways. In these instances, the comment was uncivil, but there was no evidence that the feature should be included in the dictionary.

The second reason was mathematical. If we used the comment as the coding unit, we would have needed to run reliability on each set of 25 comments. Given that the comments all included the same feature, there was a heightened possibility that many of the comments would use the feature in a similar manner. This raises problems for intercoder reliability statistics. In a situation such as ours with two coders and a dichotomous code, the formal equation for Krippendorff's alpha is $\alpha = 1 - (n-1)\frac{\text{disagreements}}{number\ of\ 0s * number\ of\ 1s}$ where n is total number of paired values (Krippendorff, 2011). Take the hypothetical situation where a feature prompted two coders to agree that 23 sampled comments used the feature in a way that aligned with the dictionary and to disagree on whether two comments used the feature in a way that aligned with the dictionary. For this case, Krippendorff's (2004) alpha is –.02—well below minimally acceptable reliability of .67. Essentially, coders could overwhelmingly agree that a feature should or should not remain in a dictionary, but still fail to achieve reliability.

For these reasons, we analyzed whether the feature should be included in the dictionary, not whether each comment was uncivil. In line with other reliability norms (e.g., inter-coder reliability is considered strong when .80 or higher), we instructed coders to include a feature in a dictionary if 80% or more of the comments used the feature in a context that matched the dictionary topic.[4]

Once all coders determined whether a feature should (coded 1) or should not (coded 0) be included in the dictionary, we ran inter-coder reliability on the include/not include decisions for all of the proposed features within the dictionary. Our reliability level was strong (Krippendorff's alpha = .87). Coders disagreed on whether a word should be included in the dictionaries only 7% of the time. We took a conservative approach and did not include any features in the dictionaries that prompted disagreement. This process pared our dictionaries down from 88 to 55 features (see Table 1). Testing whether each feature should be included in the dictionary turned out to be prudent. Features that *were not* included in the dictionary demonstrate how context can matter. We have already covered one example of a word—"devil"—that could appear on its face to involve name-calling but that actually appeared in perfectly civil contexts. Another example is the word "lynch," which could have referenced a murderous civil rights violation. Our coding procedure, however, uncovered many comments that referenced the financial institution Merrill Lynch or soldier Jessica Lynch.

The uncivil words dictionary appeared in 10% (N = 959,043) of the comments. This percentage was smaller than previous research about incivility in news comments (e.g., Coe et al., 2014), suggesting that we did not capture all incivility in the data set (although Coe and colleagues looked at a local newspaper making the comparison equivocal). However, given the human coding of the dictionary features, as well as the procedures we describe next, we are confident that the comments coded as including uncivil words are valid.

Validating the Approach

We validated the approach in two ways. First, we created a separate dictionary of swear words from the *Times* data set, which offered a unique opportunity to compare our results to words that have been consistently considered swear words over time. Across data sets of "public taboo words" collected over time, 10 words made up approximately 80% of the

Table 1
Dictionaries

New York Times Study		Chattanooga *Times Free Press* Study
Swear Words	Uncivil Words	Poverty
Included: ass, bs, crap, damn, hell	Included: dumb, farc, hypocrit, insan, insecur, irrespons, sham, trivial, unaccept, uneth, unfair, bigot, bigotri, discrimin, prejudic, racism, racist, segreg, stereotyp, betray, enemi, insurg, overthrow, riot, threat, threaten, traitor, treason, tyranni, brainwash, deceiv, decept, dishonest, disingenu, inaccur, incorrect, misinform, mislead, uninform, dysfunct, infring, obstruct, obstructionist, suppress, unconstitut, chao, debacl, delud, demean, disrespect, fiasco, inappropri, nasti, vitriol, yell	Included: employment, good-paying, income, inequality, low-income, low-wage, poverty, recession, unemployment, upward-mobility, wage
Not Included: screw, suck, swear	Not included: cheat, controversi, crimin, desper, devil, dismal, egregi, helpless, horrend, horribl, horrif, illeg, miseri, neglig, squander, terribl, unpleas, offend, endang, lynch, revolt, disenfranchis, filibust, impeach, bogu, distort, fact, irrelev, lie, brutal, meltdown, outrag, toxic	Not included: activist, Appalachian, at-risk, benefit, blue-collar, career, development, economic, economy, employee, employer, equality, expense, good-paying, hardworking, higher-wage, homeless, income, inequality, lower-income, low-income, low-wage, middle-class, mobility, poverty, problem-solve, prosperity, puzzle, recessions, reform, solution, struggling, support, underclass, unemployment, upper-class, wage, wealth, welfare, white-collar, working-class, abandon, afford, challenge, circumstance, class, community, communities, deficit, develop, disadvantaged, disconnected, earnings, economically, economist, employed, employment, finance, financial, hardship, isolated, job, opportunities, oppressed, poor, prosperous, recession, redevelopment, resources, rich, salary, shelter, stability, stabilized, struggle, upward-mobility, work, workforce

uses of swear words (Jay, 2009). This consistency suggests that swear words could be developed into an out-of-the-box dictionary. However, we created the swear words dictionary using the manually validated organic dictionary model to test whether the process could create a dictionary that compared favorably to an external construct.

Two of the authors reviewed the top-features list for all features that could be related to swearing (e.g., "damn"), leading to a list of eight features. We randomly selected 25 comments that included each feature (N = 200). Coders reached perfect reliability (Krippendorff's alpha = 1.00). The swear words dictionary appeared in 2% (N = 142,377) of the comments. Notably, every one of the features that we included in our final dictionary related to one of the words in Jay's (2009) list (see Table 1). Of the words from Jay's (2009) list that did not make ours, two were not in the top-5,000 features, two were multiword phrases that were not captured by our approach, and one ("sucks") was in our preliminary list but, because some comments used the word in unrelated ways (e.g., being sucked away from shore), it did not meet our standard for inclusion in the dictionary. Overall, although our dictionary did not capture all of the taboo words listed in Jay's (2009) work, the dictionary was validated in that all included words have been considered taboo words in previous research.

Second, to examine our method empirically, we tested a sample of 400 comments against several existing methods. To ensure variability in the comment content, we used stratified random sampling: some of the comments were identified through our method as including uncivil words (N = 75), some coded as including swear words (N = 75), and others coded as including neither (N = 250). To begin, two of the authors with extensive experience coding incivility reviewed the comments to come up with a gold standard for whether incivility and swear words were referenced. We determined that 56% of the comments included references to incivility or profanity. We then compared other measures—our manually validated organic dictionaries, two out-of-the-box sentiment dictionaries, one machine-learning algorithm, and data from two undergraduate coders trained with a codebook (see Appendix B)—to our gold standard. We combined swear words and incivility for these tests to be comparable with other methods.

This comparison is imprecise. Two out-of-the-box dictionaries (Lexicoder and Linguistic Inquiry and Word Count [LIWC]; see Pennebaker, Booth, Boyd, & Francis, 2015) test for negativity, as opposed to incivility. The machine-learning algorithm, Google's Perspective, is designed to identify comments that are "rude, disrespectful, or unreasonable."[5] This seems to encompass our definition of incivility as violations of politeness norms, but may be less connected to incivility defined as violations of democratic norms. Also, none of the three (Lexicoder, LIWC, Perspective) measure incivility as present or absent; they measure it using a range, requiring us to make decisions about cutoffs. Lexicoder returns the number of negative and positive words and their negations. We considered negative words that had not been negated and negated positive words as uncivil. In our data, this ranged from 0 to 29 (M = 4.67, SD = 5.30). LIWC returns the percentage of negative emotion words within the text. In our data, this ranged from 0 to 33.33 (M = 2.78, SD = 3.29). As we were looking for any instance of negativity, we dichotomized the Lexicoder and LIWC results into 0, meaning no negative words detected, and 1, meaning that they were detected (anything greater than 0). The cutoff for Perspective, which assigns a toxicity score from 0 to 1 (with one being most toxic) was less clear, so we evaluated what cutoff maximized the percentage correct at 0.05 intervals. A toxicity measure of 0.2 maximized the percentage correct as compared to our gold standard.

Table 2
Comparison of Methods to Gold Standard

	% Correct	% False Positive	% False Negative
Manually validated organic dictionaries	73.1	4.3	22.8
Lexicoder	68.3	30.8	1.0
LIWC	70.8	22.8	6.5
Toxicity	74.6	8.8	16.8
Trained coder #1	70.1	1.3	28.7
Trained coder #2	83.3	10.3	6.5

In order to compare these methods, we look at several data points. First, we are ultimately concerned with maximizing the percentage correctly classified. Second, we are concerned primarily with the false-positive rate (detecting incivility when, in fact, none exists). Any approach to categorizing a data set of this size for a complex construct like incivility is going to have false negatives—it is not possible to categorize every single word that will be used throughout the data set. Minimizing the false-positive rate, however, provides a way to know that the method is detecting a reliable signal.

Table 2 shows the results. The manually validated organic dictionaries correctly categorize 73.1% of the comments and yield a 4.3% false-positive rate. The out-of-the-box dictionaries are less successful; LIWC classifies 70.8% and Lexicoder 68.3%, but both frequently return false positives (LIWC = 22.8%, Lexicoder = 30.8%). The toxicity measure does slightly better in terms of the percent correct (74.6%), and slightly worse in terms of the false-positive rate (8.8%) compared to our manually validated organic dictionaries. Two trained coders fell either slightly below or above our manually validated organic dictionaries, earning percentages correct of 70.1 and 83.3. Yet the coder with the highest percentage correct also had a high rate of false positives (10.3%).

Based solely on this table, one may be tempted to conclude that the toxicity and human coding methods are similar to our manually validated organic dictionary approach. But a more nuanced evaluation is necessary. First, application of the toxicity measure involves an arbitrary cutoff. We selected the cutoff that provided the highest percentage of correct categorizations, but what that toxicity score of 0.2 means is not particularly clear. In addition, there are issues with transparency and replicability. The model has been trained on unknown data sets, beyond the initial (and publicly available) training set. Also, anyone using the Perspective application programming interface (API) can submit data that may be used to train the model further, introducing further unknown data into the training set. This means that results garnered at one time point may not be replicable at another. Essentially, the measure may be comparable, but it is a black box—manually validated organic dictionaries allow for more transparency.

The trained coders show promise, but humans cannot possibly code a data set of this size, meaning this method needs to be used in conjunction with machine learning. Note, however, that our approach does not return a data set that can be used for supervised machine learning. The unit of analysis for supervised machine learning training samples is the text, not the word. Future research should code each sampled comment twice—once for discussion of incivility anywhere in the comment, and once for whether a specific feature aligns with incivility—then compare the results of the dictionary approach to the results of a supervised machine learning approach where the training data set comments are selected based on frequently used features as detailed here.

Replicating the Approach With Issue Content

We replicated the process with a second study designed to compare the language used in the Chattanooga *Times Free Press* Pulitzer finalist poverty feature series with the language used in tweets. We first identified the most common words in the *Times Free Press* poverty series. We chose words directly from the news series so we could measure whether the Chattanooga, Tennessee, community adopted the language from the stories in their tweets after the series was published. From the top 5,000 unique words, two researchers identified 78 related to poverty. We collected all tweets geotagged in Chattanooga, Tennessee, from December 1, 2015, to June 1, 2016 (the series was published March 6) that contained any of these 78 words. For each word, a random 25 tweets were examined to determine whether the word should be included in the poverty dictionary. We identified 11 words where two trained coders (an undergraduate student and a research assistant with a BA) agreed that 80% or more of the tweets related to poverty (Krippendorff's alpha = 1.0).[6] Next, we again queried the Twitter API for Chattanooga geotagged tweets, this time only for those containing these 11 words (see Table 1). In this way, we assessed how many of the more than 5 million geotagged tweets in our original data set discussed poverty using language that mirrored the *Times Free Press* coverage (N = 4,768 tweets).

In this replication, we show several things. The first is that this method works across different types of texts. Tweets are inherently different from news comments in a variety of ways, and yet, our approach was easily implemented. Second, this test is conservative. Including only 11 terms guarantees that we underestimated the extent to which Twitter users in Chattanooga were discussing poverty. Yet, we find that the published series significantly increased the extent to which Twitter users in Chattanooga discussed poverty using the language of the news coverage in the days following the release of the news feature. Our analysis revealed that tweets about poverty in the Chattanooga area increased by nearly 28% for one week following the publication of "The Poverty Puzzle Series" by the Chattanooga *Times Free Press*. There are other ways to conduct this type of analysis; through the manually validated organic dictionaries approach, however, we were able to tie the language in tweets directly to language used by the news organization.

Discussion

By no means do we suggest that the method outlined in this manuscript solves all of the problems facing large-scale content analyses. However, it offers researchers another tool they can use when analyzing content from large data sets, particularly when they want to conduct a deductive content analysis of context-dependent content. Based on our experience with this approach, we make the following recommendations for researchers.

Recommendations

Create Dictionaries From Content to Account for Context. Although out-of-the-box dictionary approaches have the benefit of consistency across studies, those dictionaries do not work well when (a) they do not have a strong conceptual fit with a concept of interest and (b) the words related to that concept vary across contexts. Our manually validated organic dictionary approach is helpful in these situations. For instance, the dictionary we created for uncivil words when analyzing the *Times* comments data set included the feature "chao." In 2018, if we included that feature, comments could mention United States Secretary of the Treasury Elaine Chao, indicating that "chao" no longer relates as strongly to incivility. Since the final comments in our data set were posted in 2013, this conflict was not a problem. By asking coders to validate the specific word use and by drawing those words from the study context, the manually validated organic dictionary approach offers the opportunity to capture terms that may have varying meanings.

Don't Expect to Capture Everything, but Make Sure What You Do Capture Is Valid. None of the dictionaries in our studies captured all of the uncivil or topical content present—nor, we submit, would any of the presently available approaches. We demonstrated that the manually validated organic dictionary approach compared favorably to sentiment analysis, Google's Perspective API, and trained human coding. Although there certainly was incivility and, for the Chattanooga study, poverty discussion, missed by our method, the technique is helpful for predicting relationships among types of content (e.g., news and tweets) and among content and other variables (e.g., recommendations of news comments).

Larger Dictionaries Are Not Always Better or Necessary. Notably, our dictionaries contained far fewer words than typical out-of-the-box dictionaries. Large dictionaries are not always necessary, but it is necessary that the content in the dictionary theoretically aligns with the concept being measured. For instance, our swearing dictionary included only five features, and yet still closely matched other lists of swear words, appeared in about 2% of the *Times* comments, and predicted significant effects in our analysis (see Muddiman & Stroud, 2017). It is important to be precise and confident that the terms in the dictionary match the concept.

Future Directions

The method described in this article should prompt further research both to sharpen the details of this process and to expand its reach. First, some of the details of the manually validated organic dictionaries method would benefit from testing. Does a starting list of 1,000, 5,000, or 10,000 features yield the dictionary that best balances correct categorization, minimal false positives, and workload? How many texts that include a specific feature should be sampled to minimize false positives? And what percentage of texts must use a feature in a manner that aligns with a dictionary to be included in that dictionary—80% or something higher? Also, more research should be conducted to look into ways that large-scale data analysis methods could be used in conjunction with one another.

Future projects can use this approach in other contexts and on other platforms. Partisan language, for example, likely changes over time. The example of Secretary Chao just detailed demonstrates the possibilities for partisan content—the word "chao" should appear in a partisan dictionary post-2017 rather than an incivility dictionary. In addition, researchers should test the manually validated organic dictionary approach across different social media and with other textual materials. By starting with a list of features posted to specific platforms with specific affordances, researchers can make sure they capture content that is unique to those spaces.

Conclusion

We provide a new method for large-scale content analysis that allows political communication scholars to keep control of what we know best—language. In circumstances where a theoretical concept is context-dependent, we suggest that creating dictionaries from the context under investigation and manually validating each feature included in a dictionary can help scholars balance the need for theoretical control over computer categorization of content and the need for flexible analyses that take context into account. Political communication researchers have been approaching "text as data" for decades. We should harness our expertise by continually returning to and validating the language itself.

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Supplementary material

Supplemental data for this article can be accessed on the publisher's website at 10.1080/10584609.2018.1517843.

Notes

1. We received the data set directly from the *Times*. It includes the population of English-language comments (N = 9,616,211) posted between October 2007 and August 2013. The *Times* conducts pre-moderation on nearly all comments. We had access to comments that were both accepted to and rejected from the site.
2. The Porter stemmer has become the standard method for stemming English-language content (Diakopoulos, 2015; Willett, 2006).
3. The stemming process "tends to improve the performance of information retrieval," because fewer words need to be scanned (Jurafsky & Martin, 2016, p. 68). Although we stemmed, we note that there is some discussion about whether stemming is always appropriate (e.g., Schwartz & Ungar, 2015).

4. Since the 80% cutoff (20 out of the 25 comments) was arbitrary, we conducted a robustness check with dictionaries pared down to include only features where both coders agreed that 24 of the 25 comments used the feature correctly. This approach decreased the percentage of the *Times* comments that included uncivil words (from 10% to 6%), but the analysis patterns we uncovered remained consistent (see Muddiman & Stroud, 2017). This suggests that a stricter cutoff may not be necessary, but more work needs to be done to test for the best cutoff.

5. See https://www.perspectiveapi.com/.

6. The two coders were asked whether a tweet mentioned each feature in the context of poverty, coding 1 if yes and 0 if no.

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